

Yuchen Wang

Agentic post-training | ML systems

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Education

University of Illinois Urbana-Champaign	2025 – 2027 (expected)
Master of Computer Science	Champaign, IL
Peking University	2021 – 2025
B.S. in Intelligent Science and Technology (AI); Zhi Class (Elite Program), Top 10%	Beijing, China

Internships

Alibaba U.S. – Accio Team	May 2026 – Dec 2026
Research Intern, LLM Training Algorithms & Infrastructure	Sunnyvale, CA
- Core contributor to Occamy-1.0 [arXiv] across agentic post-training algorithms, training infrastructure, and data; the 35B-A3B agent has 500K+ 📄 downloads across Occamy variants. Captured exact model-boundary token IDs and verifier outcomes across OpenClaw, Hermes, and Accio Work. - Recovered long-horizon SFT/RL supervision through workspace replay and context–action–reward realignment after tool failures and compaction, keeping trajectories usable for post-training. - Shipped Occamy with Accio; versus its Qwen3.6 base, the released model raised AutomationBench strict pass 7.5% → 27.6% and Claw-Eval average 69.5 → 82.2, while reducing Claw-Eval T/C full-trial time 36.6%.	
Freedo Technology	May 2025 – Aug 2025
Research Intern, 3D Vision and Generative AI	Beijing, China
Raised point-cloud geometric fidelity from 32.7% to 85.3% by conditioning ControlNet on depth and surface normals while preserving edges and planar surfaces.	

Open Source

- Halved 120B MoE TP2 BF16 conversion peak (**7.88 → 3.94 GiB**) in **vLLM**; fixed calibrated FP8 KV prefill in **FlashInfer**.
- Preserved source sharding in **HF Datasets** streaming shuffle, restoring four-worker use on 19 shards; made packed QSA selection document-local in **mcCore-bridge**, **2.70× faster** on an 8K synthetic selector case.
- Developing **Bridge-local Qwen4-Exp** support for QSA, per-layer embeddings, gated residuals, and HF conversion.
- Bypassed the Ray driver for **NeMo RL** top-*k* teacher transfers (**4.44–5.31×** on 8×B200); added **GDN kernel-fused Q/K normalization** and **selective Mamba recompute** in Megatron-LM.
- Fixed silent small-gradient failure in **Muon**; proposed **checkpoint-safe Adam counters** for Megatron-LM; co-authored a **NeMo Gym** fix preserving HTTP errors.

Research

RareDx: Graph-Grounded RL for Rare-Disease Diagnosis [arXiv]	Jun 2026 – Sep 2026
Co-first author; Advisor: Prof. Tianyu Liu, Tsinghua University	
Co-designed RL rewards for partial diagnostic matches using disease-ontology distance, biomedical similarity, and phenotype–disease links. With SFT and retrieval, 27B diagnosis Hit@10 increased from 28.53% to 40.76%.	
CineFlow: Dependency-Driven Parallel Video Generation [Paper]	Oct 2025 – May 2026
Co-first author; Advisor: Prof. Fan Lai, University of Illinois Urbana-Champaign	
- Developed a training-free, critical-path scheduler for video diffusion using semantic DAGs, trajectory fusion, and online edge pruning; analyzed 1K+ MovieGenBench prompts. - Delivered 1.7–5.5× end-to-end speedup, 1.30–2.02× lower P90 latency, and 5.4–17.3% higher VBench overall across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100.	
Dynamic Prefill Optimization via Adaptive Online Packing [Report]	Aug 2025 – Jan 2026
University of Illinois Urbana-Champaign	
Built an AIMD batch-size controller driven by live p95 TTFT and packed mixed-length prompts with greedy/DP scheduling; cut TTFT by up to 20% on production-style DynamoLLM traces.	
FlashAttention-style CUDA Kernels for GPT-2 [Report]	Aug 2025 – Dec 2025
University of Illinois Urbana-Champaign	
Accelerated GPT-2 through tiled attention, online softmax, and kernel fusion, raising arithmetic intensity from 15 to 157 FLOP/byte, cutting HBM traffic roughly 10×, and improving end-to-end speed up to 9%.	
Reinforcement Learning for Legal Reasoning [Thesis]	Jan 2025 – Mar 2025
Advisor: Prof. Yansong Feng, Peking University	
Post-trained a legal model with Zero-RL, chain-of-thought-distilled SFT, and GRPO; reached 57.6% accuracy, beating SFT-only and RL-only baselines.	

Service, Awards & Technical Skills

Reviewer, **WSDM 2027**

Awards: Merit Student; Zhi Class Scholarship; CMO (Provincial First Prize), 2020.

Languages: Python, C++, CUDA. **Systems:** PyTorch, Ray, FSDP, Megatron-LM, NeMo RL, vLLM, SGLang, Nsight.

Methods: RL, distributed training, diffusion transformers, ControlNet, 3D vision and point clouds.